Supplementary Table 1: Correspondence between parameters named in the manuscript and variables in the associated code (publicly available on GitHub).

Description	Paper	Code
Recorded density	$D_{p,t}$	dens
SD of density at the log-scale	σ_{obs}	sigma_obs
Natural mortality rate	est_m	m
Fishing mortality rate	$f_{a,t}$	f
Intercept for detection probability	β_0	beta_obs_int
Coefficient linking detection probability to density	β_1	beta_obs
Overall average recruitment	μ	mean_recruits
Conditional variance in the AR1 process	σ_{proc}	sigma_r
White noise used in the AR1 process	z_t	raw
Autcorrelation from AR1 term	α	alpha
Excess in natural mortality due to temperature	γ	m_e
Isotropic dispersal rate	δ	d
Sea bottom temperature	$T_{p,t}$	sbt
How much tax per unit of temperature	β_{tax}	beta_t
Suitability index	$I_{p,t}$	T_adjust
Temperature optimizing $I_{p,t}$	τ	Topt
Width parameter from $I_{p,t}$	ω	width

Supplementary Table 2: Hyperparameters used in the DRM.

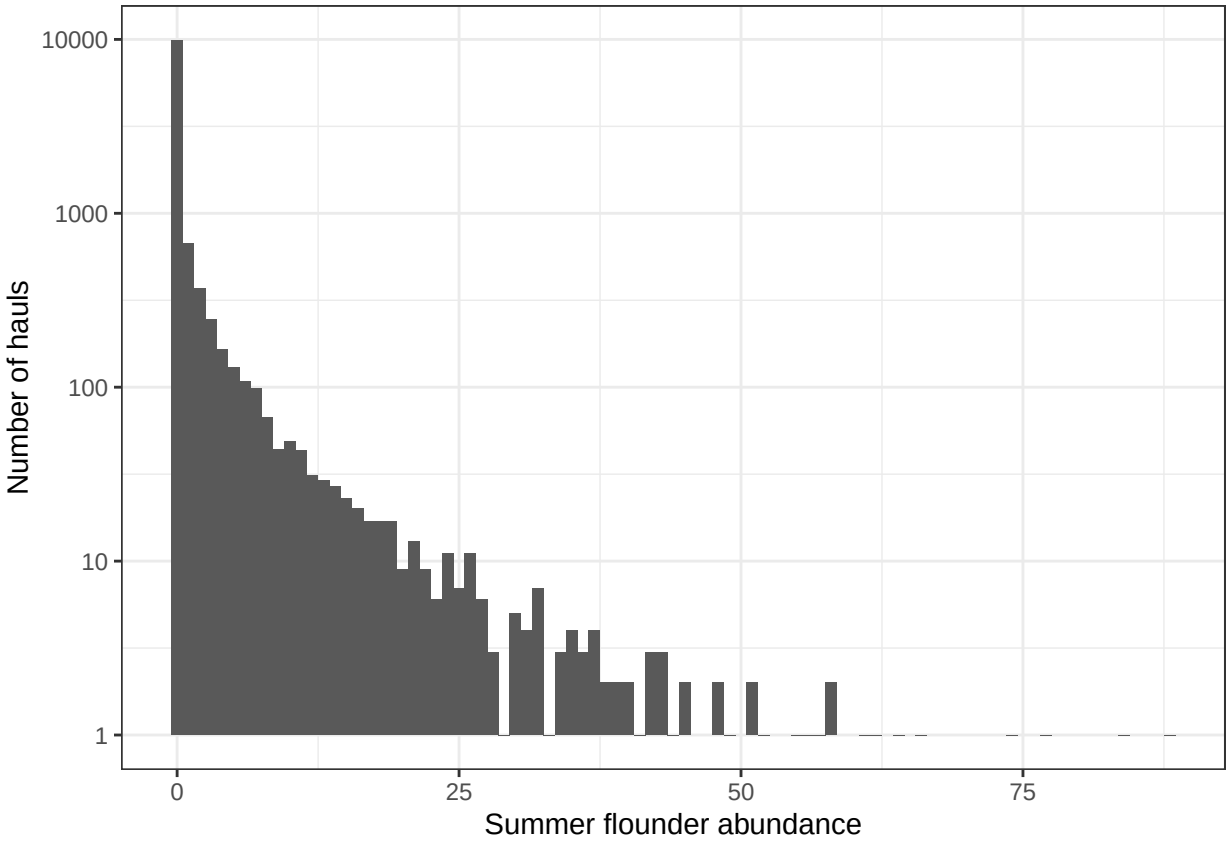
Parameter	Prior.Distribution
$\beta_{obs,1}$	Normal $\sim (0.001, 0.1)$
$\beta_{obs,0}$	Normal $\sim (-100, 4)$
z	Normal $\sim (0, 1)$
σ_{proc}	Normal $\sim (0.2, 0.1)$
σ_{obs}	Normal $\sim (0.2, 0.2)$
$width$	Normal $\sim (4, 2)$
T_{opt}	Normal $\sim (18, 2)$
d	Normal $\sim (0.01, 0.1)$
β_{tax}	Normal $\sim (0, 2)$
α	Beta $\sim (12, 20)$
μ_r	Lognormal $\sim (7, 5)$
m_e	Exponential ~ 2.3

Supplementary Table 3: Bayesian linear regressions of sea bottom temperature (measured in the trawl surveys) on year within each patch during the model training interval (1972-2006) with Gaussian errors, summarized by the median and 95% Highest Posterior Density Interval (HPDI) of the posterior estimate for the slope term. Values are truncated at three significant digits.

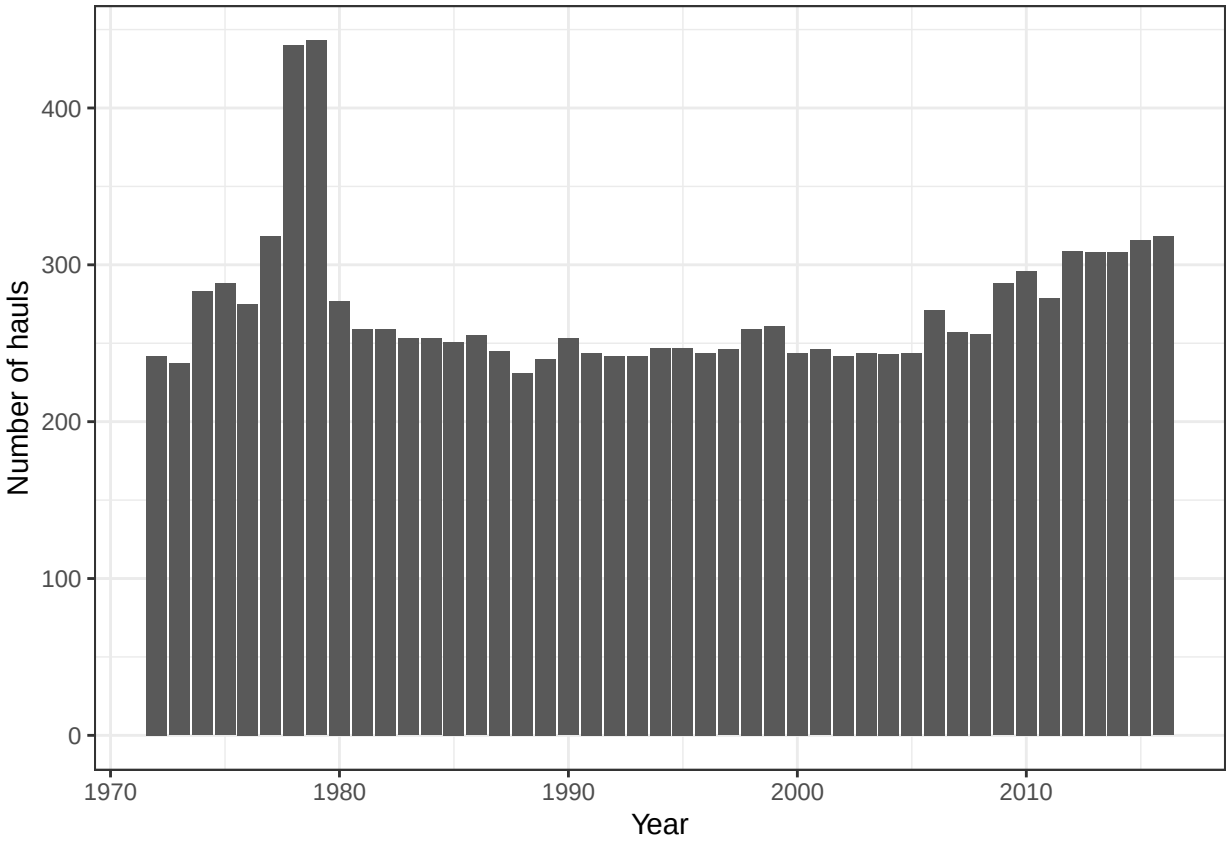
Patch	Median	HPDI
35	0.00236	-0.0734 — 0.0803
36	0.04540	0.00128 — 0.0909
37	0.05610	0.0106 — 0.0972
38	0.02900	-0.00582 — 0.0654
39	0.04710	0.0203 — 0.0709
40	0.01940	0.00837 — 0.0309
41	0.02680	0.0108 — 0.0428
42	-0.00264	-0.0103 — 0.00479
43	0.00328	-0.00566 — 0.0123
44	0.00607	-0.0154 — 0.0253

Supplementary Table 4: Bayesian linear regressions of sea bottom temperature (measured in the trawl surveys) on year within each patch during the model testing interval (2007-2016) with Gaussian errors, summarized by the median and 95% Highest Posterior Density Interval (HPDI) of the posterior estimate for the slope term. Values are truncated at three significant digits.

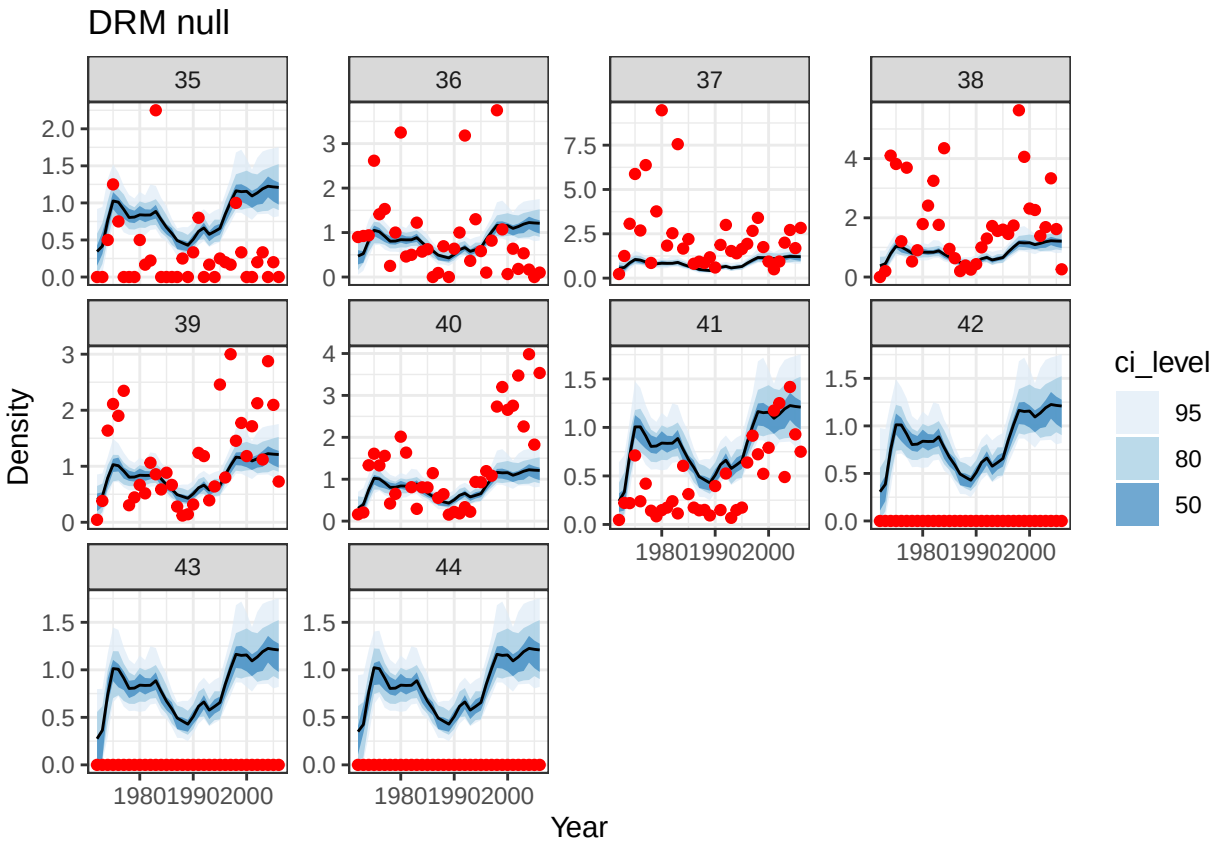
Patch	Median	HPDI
35	0.1520	-0.31 — 0.598
36	0.2310	-0.139 — 0.561
37	0.1620	-0.134 — 0.477
38	-0.0439	-0.266 — 0.17
39	0.0890	-0.0928 — 0.261
40	0.3610	0.276 — 0.446
41	0.1590	0.0441 — 0.278
42	0.1690	0.114 — 0.228
43	0.1740	0.106 — 0.243
44	0.2530	0.159 — 0.342



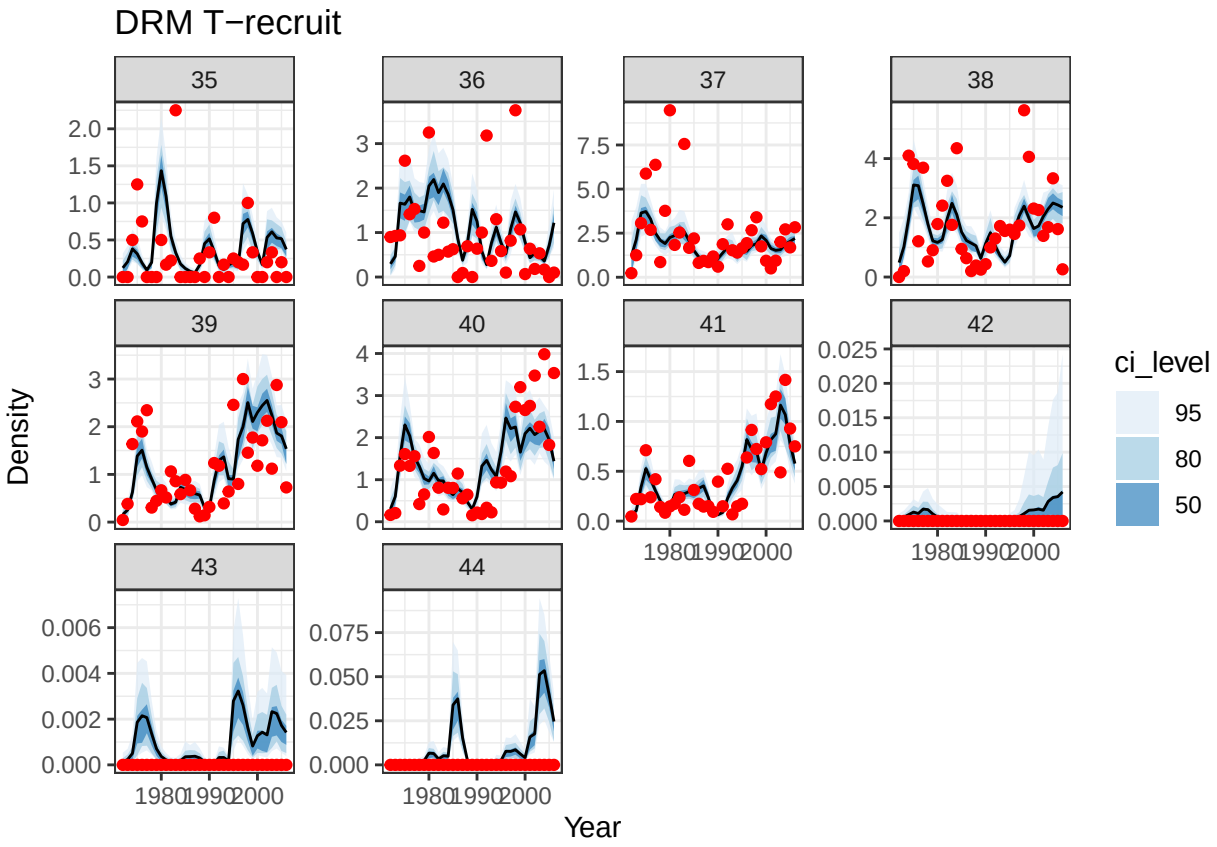
Supplementary Figure 1: Frequency distribution of summer flounder abundance in all hauls used in the analysis (testing and training data combined; n = 12,203).



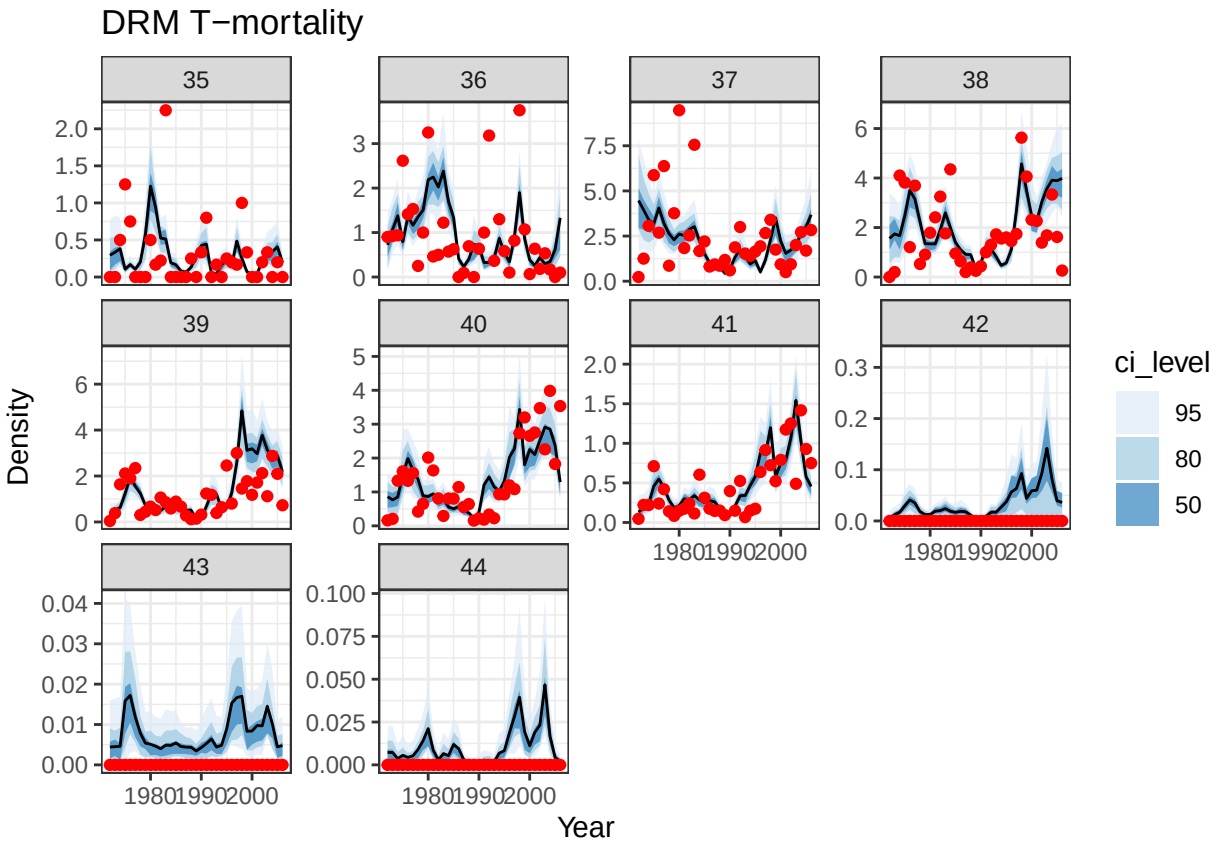
Supplementary Figure 2: Number of hauls per year; n = 12,203.



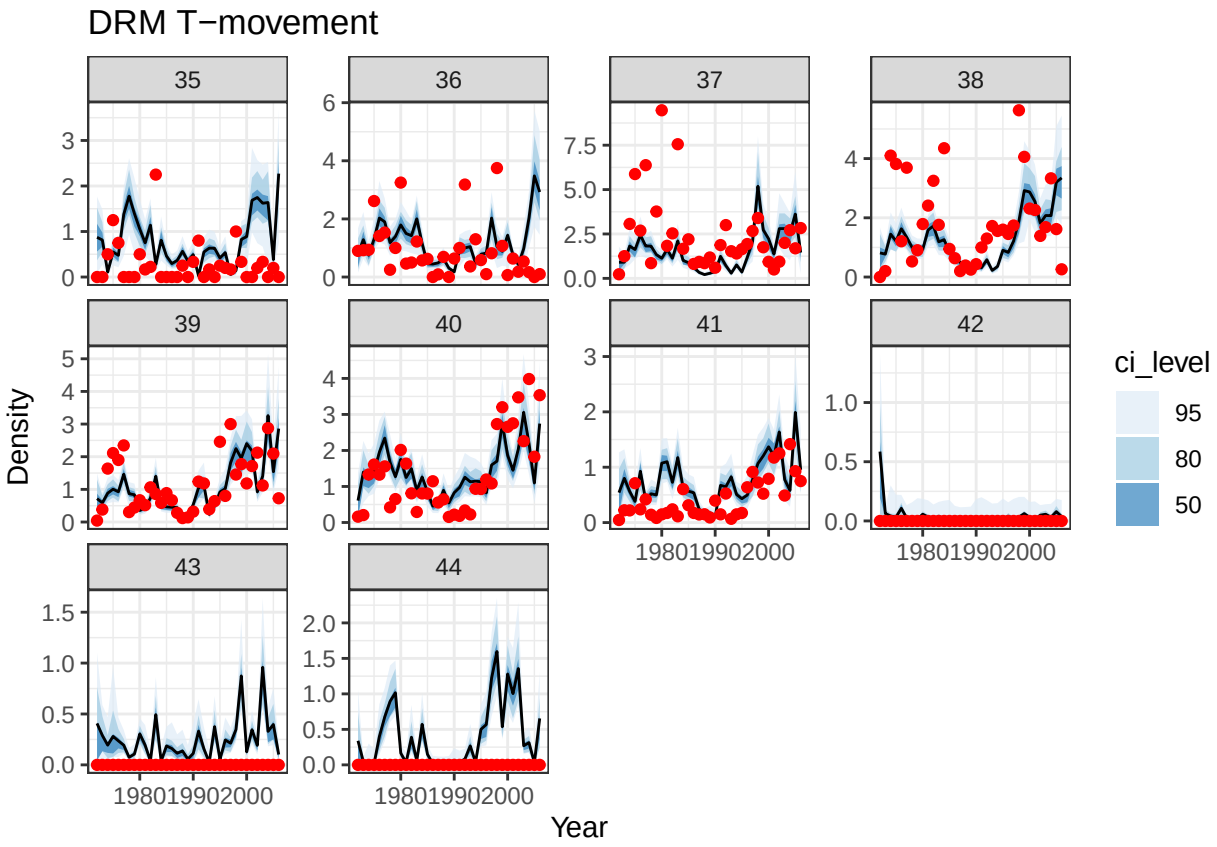
Supplementary Figure 3: Median density estimated by the null DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



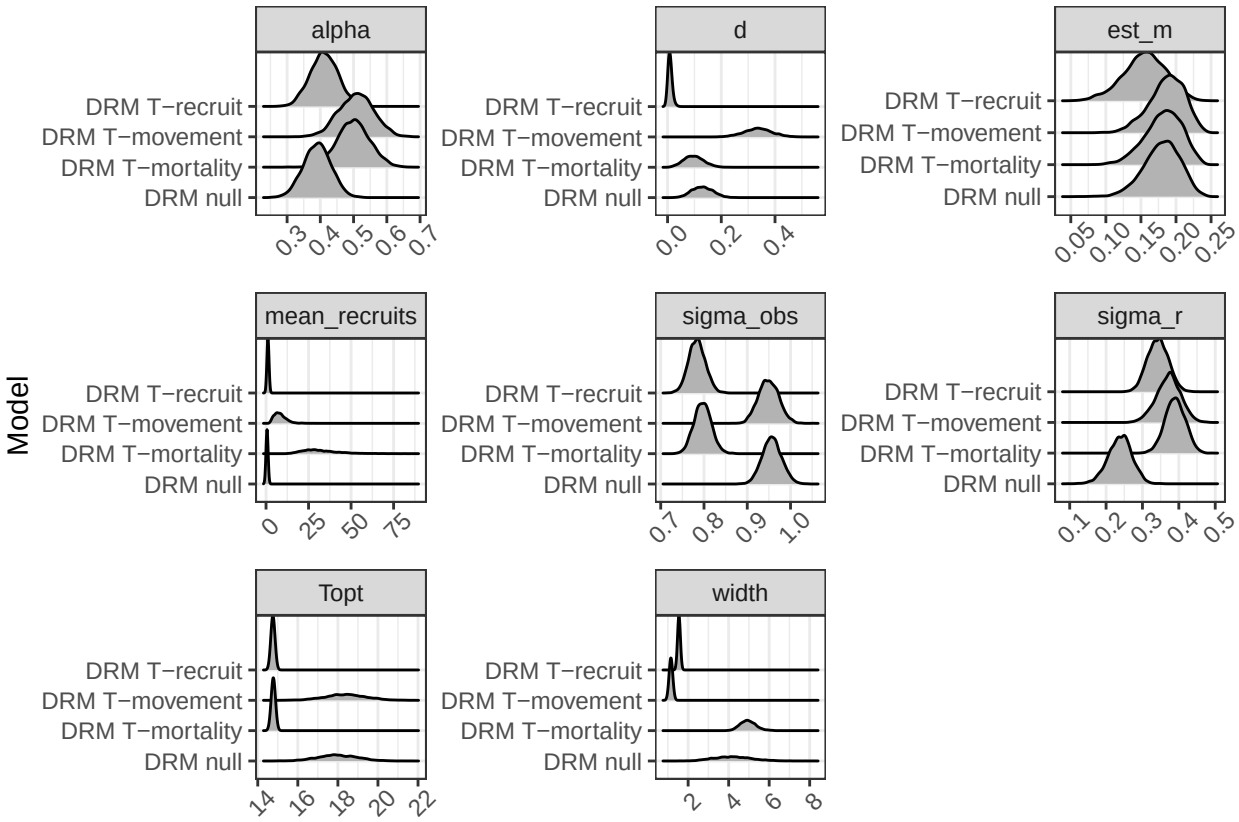
Supplementary Figure 4: Median density estimated by the temperature-dependent recruitment DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



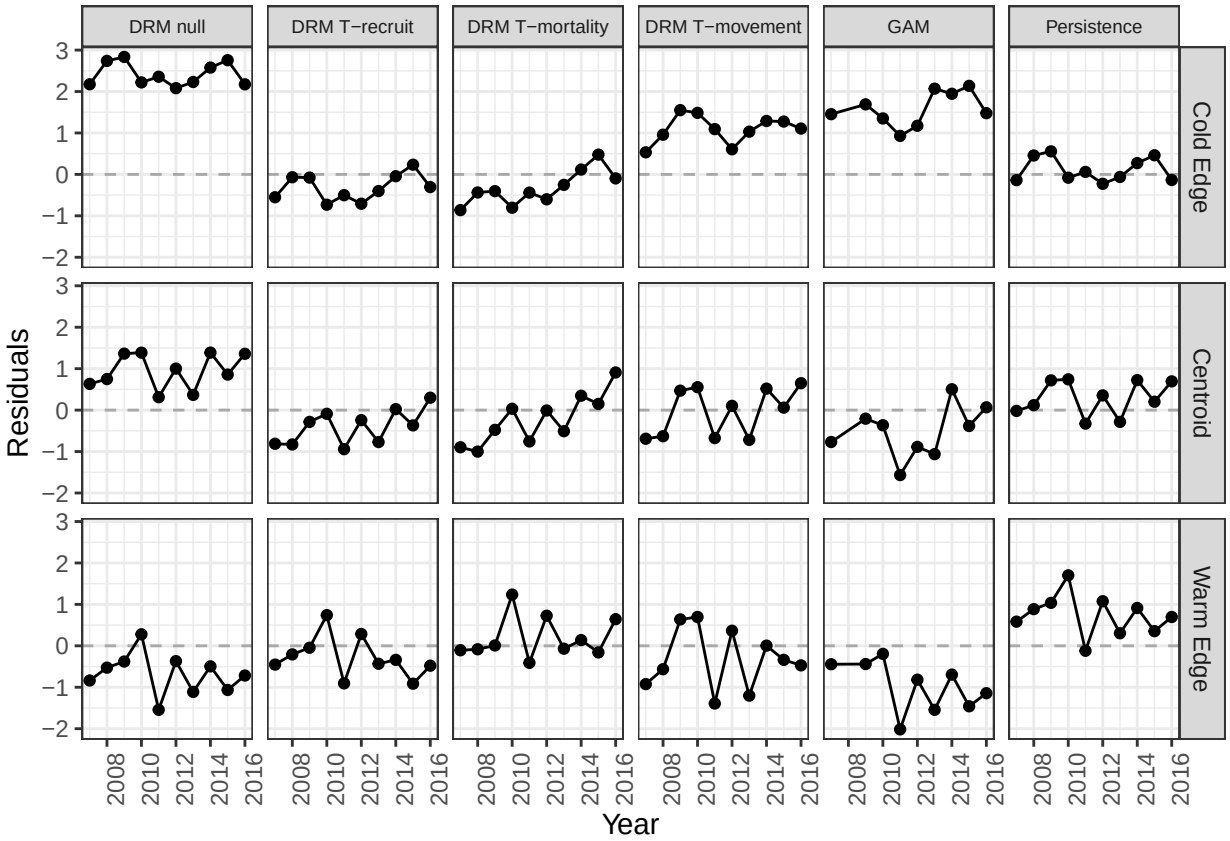
Supplementary Figure 5: Median density estimated by the temperature-dependent mortality DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



Supplementary Figure 6: Median density estimated by the temperature-dependent movement DRM in the training dataset by patch and year (black line). Blue shading represents the 50%, 80%, and 95% credible intervals. Red points are the observed data. Note that y-axes vary by patch.



Supplementary Figure 7: Posterior distributions of parameters from the four fitted DRMs. These parameters are d, the annual dispersal fraction between adjacent patches; mean_recruits, the average density of recruits per patch; sigma_obs, the standard deviation of density at the log scale; Topt, the optimal temperature estimated for whichever process (movement or recruitment or mortality) was estimated as temperature-dependent, in °C; and width, a parameter controlling how sensitive the modeled process was to temperature. See Supp. Tab. 1 for correspondence of code variables to model parameters in the manuscript.



Supplementary Figure 8: Residuals (estimated minus observed values) of range metrics each year over the ten-year testing period for the focal models in the main text: DRM with no temperature effect (DRM null) or a temperature effect on recruitment, mortality, or movement; a GAM SDM; and a persistence forecast. The horizontal dashed line indicates zero residuals, i.e., the estimate perfectly matches the testing data.